

TEAM

TEAM NAME

SYNTAX TERROR (TAN)

Project: Parity, MCX gold futures intelligence
Problem 03: Commodity Derivatives Intelligence

Working prototype | 710 contracts | 3 sealed tests

MEMBERS AND ROLES

Nikhil Kumar
Team lead · data pipeline, strategy research and backtests, dashboard

Sai Ganesh
Market research and pitch presentation

Ishaan Chhabra
Testing, data checks and documentation

PROBLEM STATEMENT & THEME

One metal, four prices. Which one is fair?

GOLDM

100 g bar · 99.5% pure

₹1,49,055 per 10 g

₹14,980 per pure g

GOLDTEN

10 g bar · 99.9% pure

₹1,49,436 per 10 g

₹14,959 per pure g

GOLDGUINEA

8 g coin · 99.9% pure

₹1,19,993 per 8 g

₹15,014 per pure g

GOLDPETAL

1 g · 99.9% pure

₹15,002 per 1 g

₹15,017 per pure g

Most traded contract of each type, MCX closing prices, 1 October 2026. Different quotes, nearly the same gold price.

PROBLEM

- Quotes differ by size and purity, so they cannot be compared at a glance
- Is a price gap real, or just a unit mix-up?

IMPACT

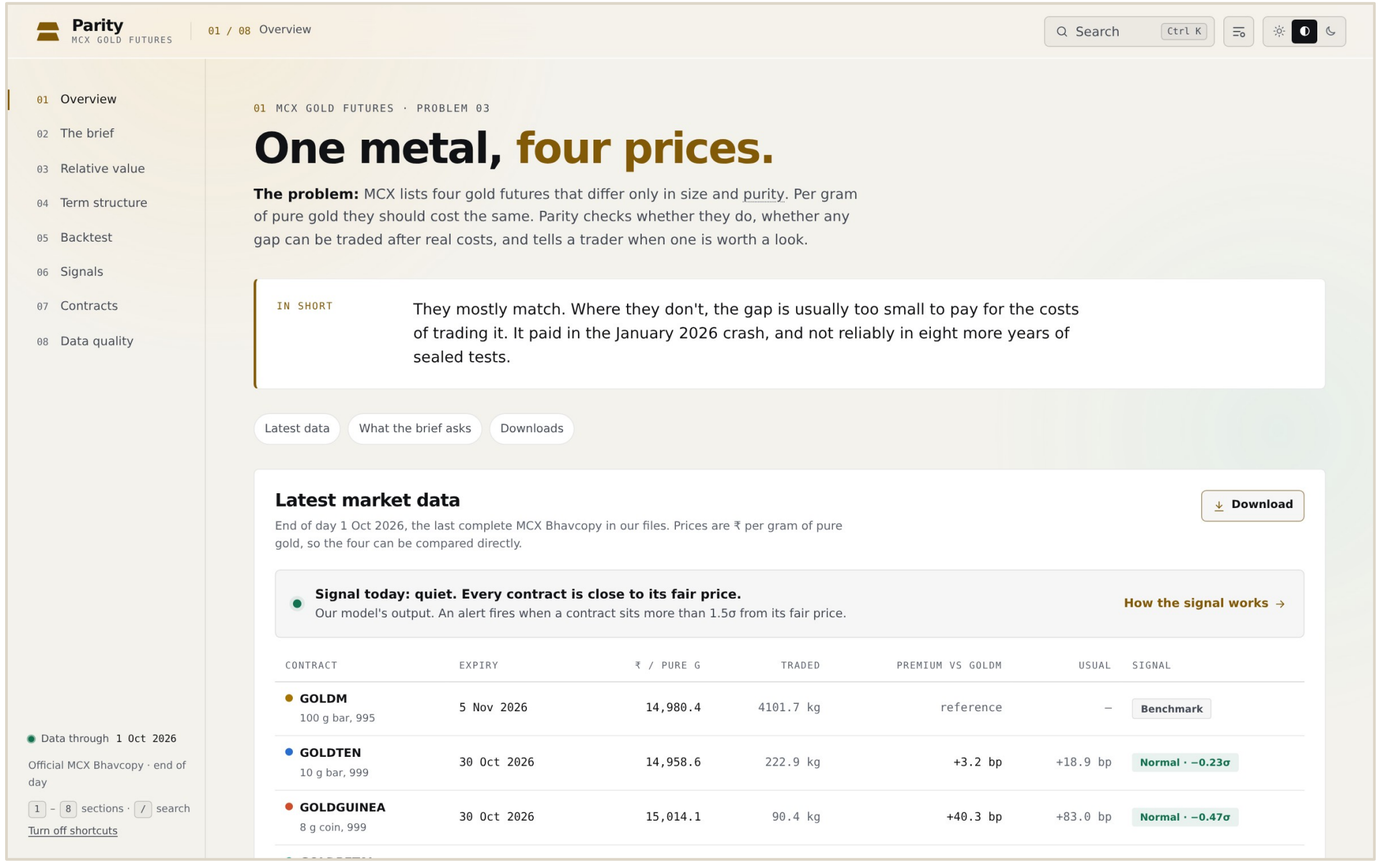
- The 8 g coin costs 0.6–0.7% more per gram, in all 41 contract months since late 2023
- In the 2026 gold crash, small contracts sat 2.7% above GOLDM

THEME ALIGNMENT

- Problem 03: Commodity Derivatives Intelligence
- Covers fair value, futures curve, backtest, alerts and contract calendar

SOLUTION

Parity tells a trader when a price gap is real



WORKING PROTOTYPE 8 pages, search and downloads, runs offline as one file

Fair comparison
All four contracts in ₹ per gram of pure gold

Quiet alerts
Speaks up only when a contract strays far from its fair price

Honest backtests
After every MCX fee and tax, on data the rules never saw

Contract calendar
Every trade checked against delivery dates

UNIQUE VALUE

Honest by design: rules locked in Git before every test, 3 sealed tests run once, and losses shown too.

ARCHITECTURE

From exchange files to a trader's screen

MCX files
Official daily Bhavcopy, 2003–2026

→

Clean & check
Duplicates out, every row checked

→

Analyse
Fair value, curve, backtests, alerts

→

Results
Every number saved to a file

→

Export
One data file feeds every output

Dashboard for traders

Report 17-page PDF

Pitch interactive story

TECH STACK

- Python, pandas, NumPy
- React 19 built with esbuild
- ReportLab and Matplotlib

HOW WE KEEP IT HONEST

- Rules time-stamped in Git
- Sealed years kept in a separate file
- 10 automated tests

APIS USED

- No paid APIs, no keys
- MCX's public Bhavcopy page
- Works fully offline

Code: github.com/nikhilkumar-dev0310/HACKINHILLS-SYNTAXTERROR-TAN-

MARKET RESEARCH / ANALYSIS

Gold futures in India: big, and growing fast

₹28,484 cr

traded every day in MCX gold futures, FY2025-26

3.4×

growth in one year (₹8,449 cr a day in FY2024-25)

57%

of all MCX trading is bullion, Q2 FY26 (44% a year earlier)

Sources: Business Today, 14 Aug 2026 (MCX data for FY2025-26); Business Standard, 7 Nov 2025 (MCX Q2 FY26 results).

TARGET USERS

- Traders choosing which gold contract to trade
- Arbitrage desks, jewellers hedging, broker research teams

MARKET SIZE

- MCX trades ₹4.11 lakh crore a day across all contracts (Q2 FY26)
- Gold comes in 1 kg, 100 g, 10 g, 8 g and 1 g lots

REVENUE MODEL (OUR PLAN)

- Free dashboard; paid alerts by email or Telegram
- Licence for broker research desks, and a data feed

EXPECTED GROWTH

- Small lots (10 g, 1 g) are bringing in new traders
- Same method extends to silver: SILVERM, SILVERMIC

FUTURE SCOPE

What comes next

01

Live paper trading

Run the fair-price rule every day on new MCX files, before any real money

02

Intraday prices

Measure real trading costs instead of assuming them

03

Alerts on your phone

Email and Telegram alerts that show the cost of acting

04

Longer history

Measure the coin and 1 g premiums back to 2008

05

Safer rules

Skip days stuck at the price limit; cross-check with spot gold rates

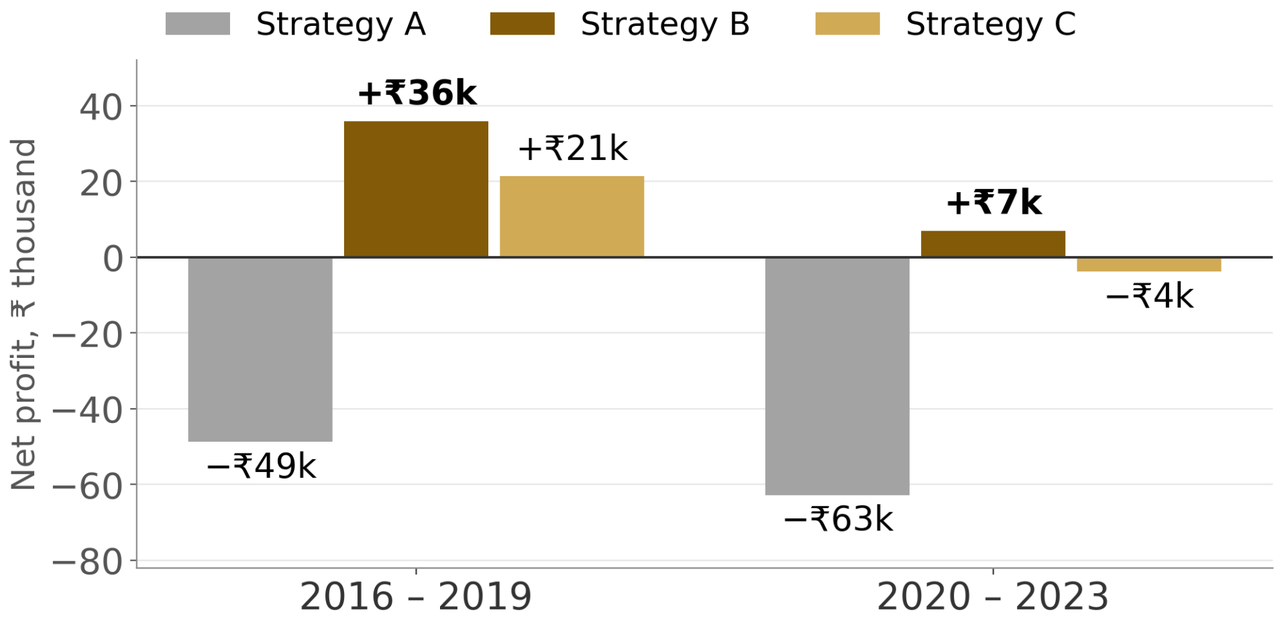
06

More metals

The same method for silver and other multi-size contracts

EXTRA: RESULTS

Tested on 8 years it had never seen



Strategy A fails.
It lost in both windows, even with perfect fills.

Strategy B holds up.
It made money in all 3 sealed tests: ₹12,990, ₹35,873 and ₹6,962.

We say it plainly.
The edge is small, and trading costs decide it.

Net profit after all costs, 5 bp slippage. Each window run once, with rules frozen first.

710	55,457	0	3
MCX contracts, 2003–2026	daily rows checked	unexplained data errors	sealed tests, each run once



Thank you!

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Nikhil Kumar · Sai Ganesh · Ishaan Chhabra

Code on GitHub